PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Kwanele Mbambo
Program/Batch	B084-Networkwalks
Date	13 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (Footp with GHDB) W2-PM3 (Footp with maltego) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5), google hacking dabase uncovering live exposed cameras(W2-PM2), Finding email addresses related to networkwalks.com (W2-PM3). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All commands were run in Kali Linux (footprinting) AND Zenmap Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification
Maltego	Finding all related email addresses of networkwalks.com
Google hacking database	Finding all exposed live cameras on internet.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.0.4** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

I installed maltego on kali linux, set it up, and searched the email associated with netowrkwalks.co.za,

I also used google hacking database to find all exposed live cameras found online using dockers and also find the mathematics exposed downloadable pdf's.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The results identified 3 live hosts:

- 10.0.0.1
- 10.0.0.2
- 10.0.0.3
-

The example results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

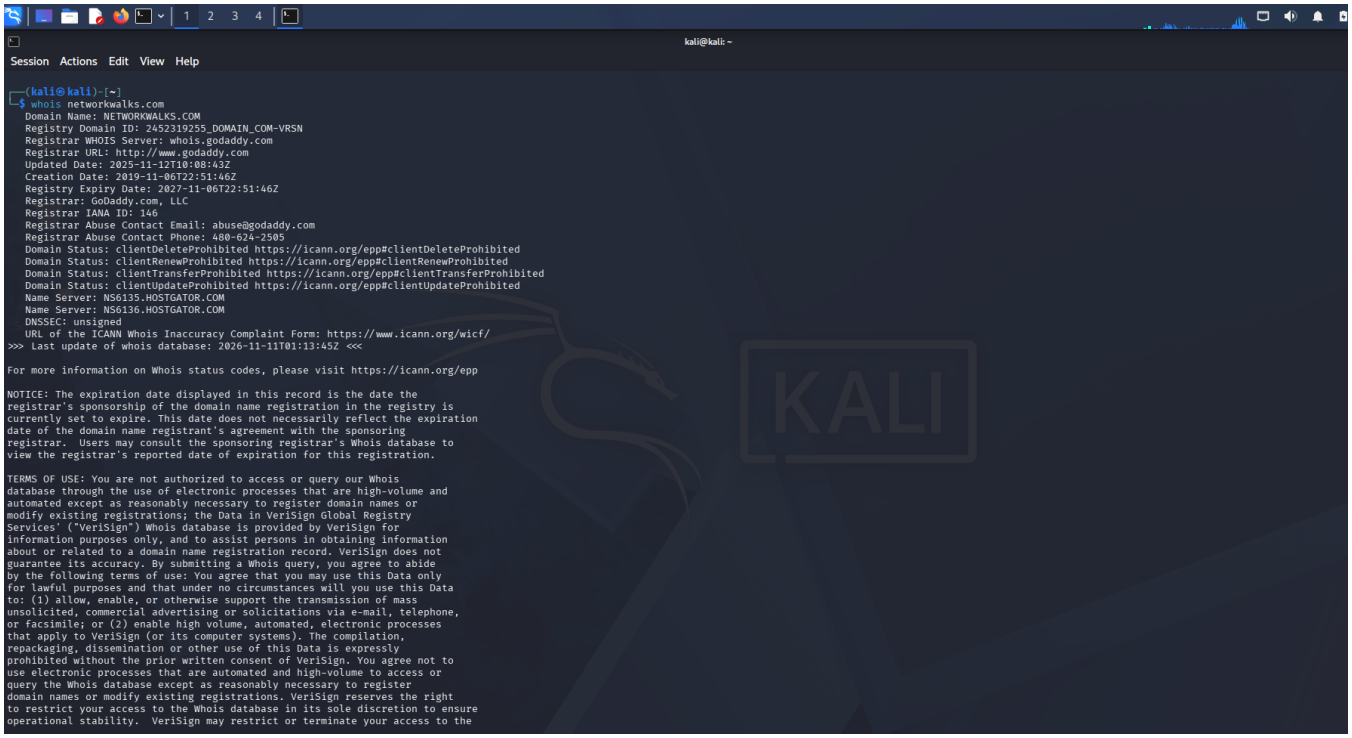
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

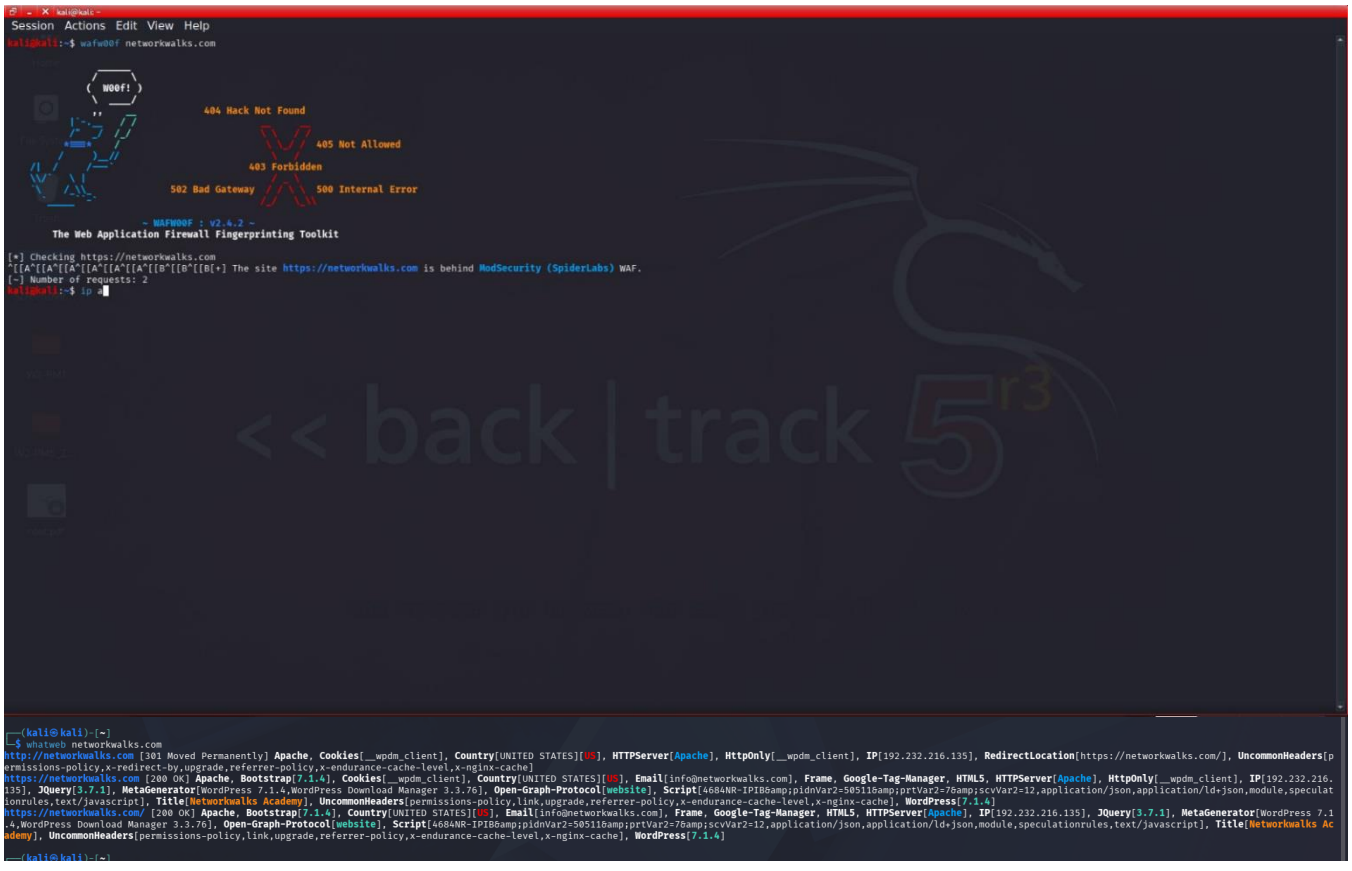
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

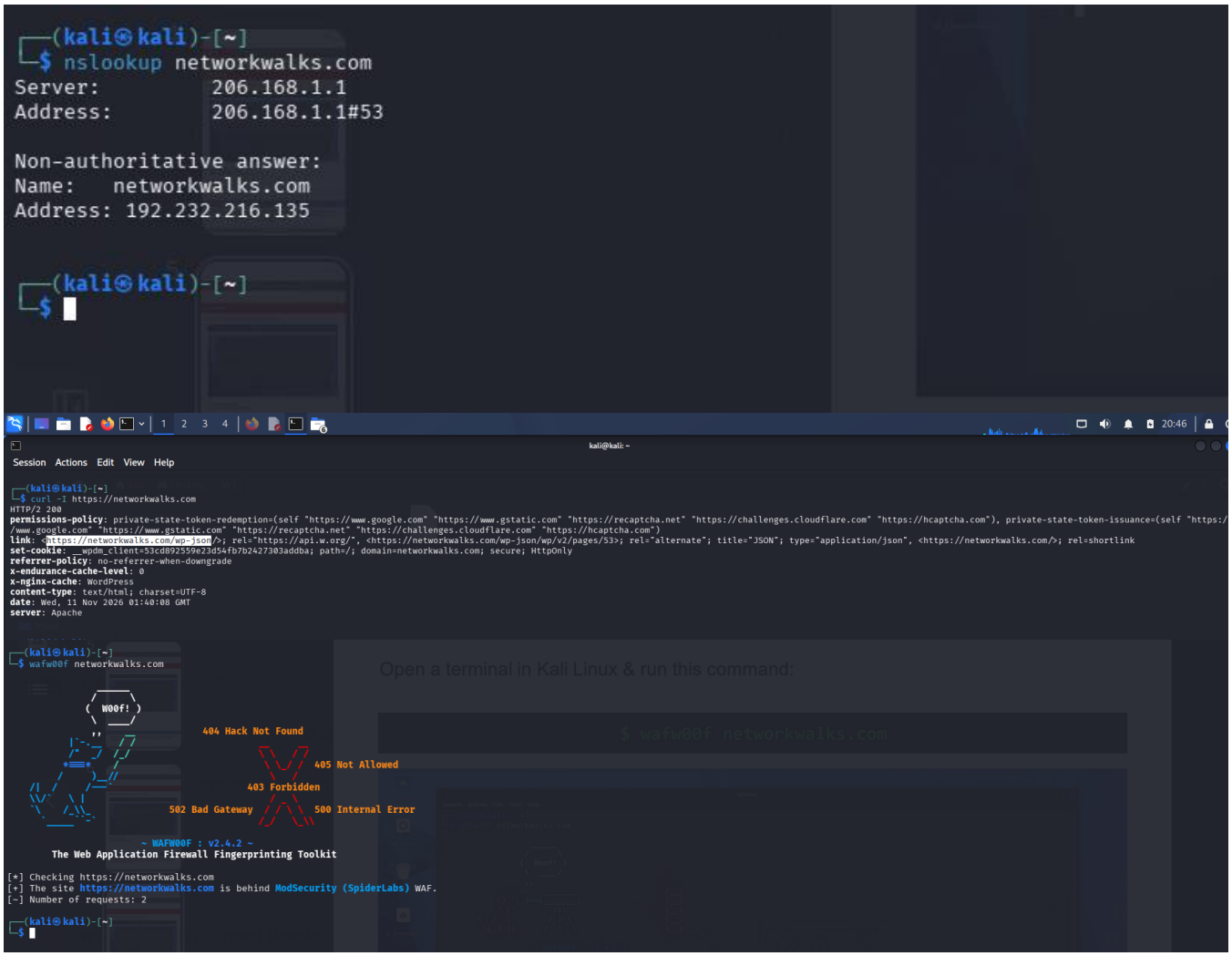
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected



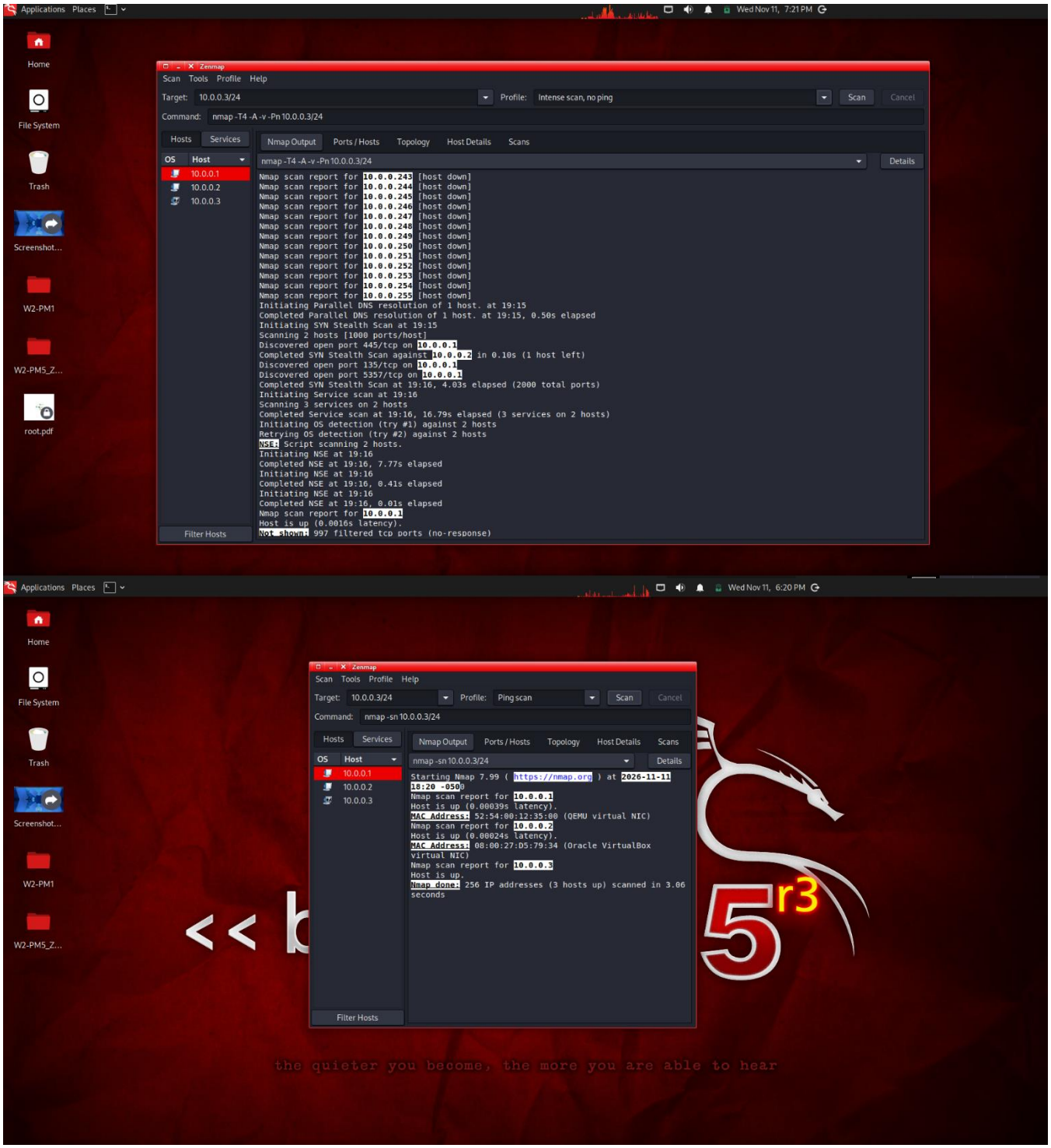


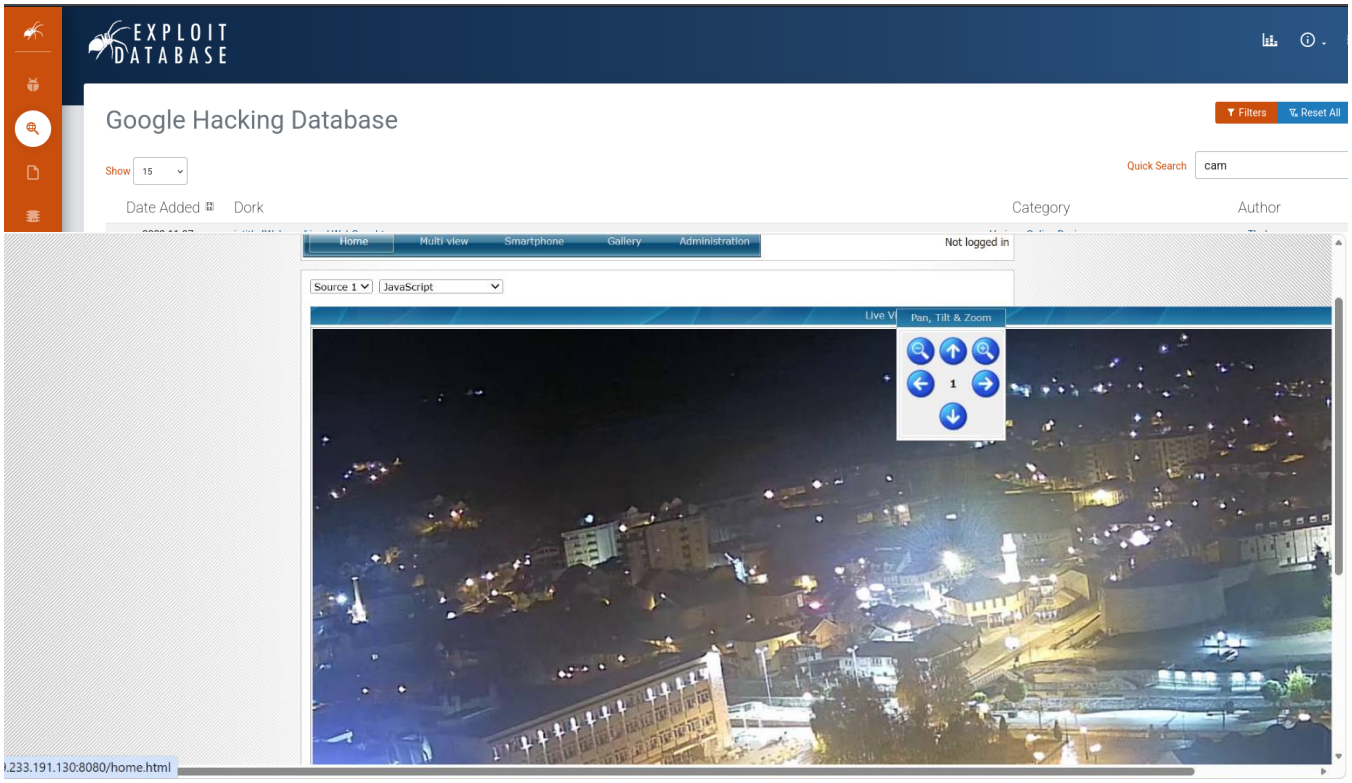

(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server: 206.168.1.1
Address: 206.168.1.1#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
(kali㉿kali)-[~]
$
Session Actions Edit View Help
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=53cd892559e23d54fb7b2427303addba; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 11 Nov 2026 01:40:08 GMT
server: Apache
$ wafw00f networkwalks.com
W00f!
404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
Open a terminal in Kali Linux & run this command:
$ wafw00f networkwalks.com

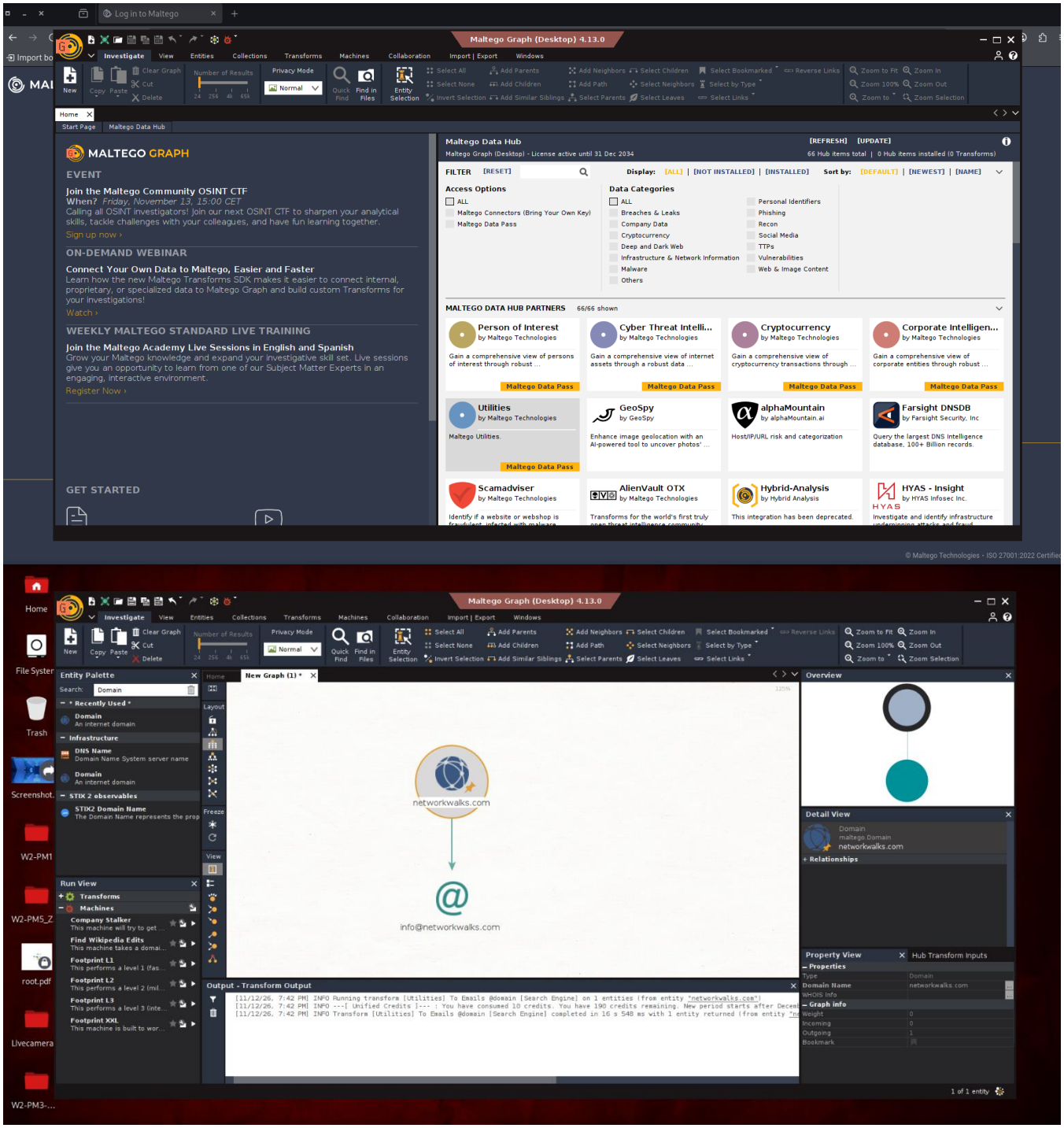

Home
File System
Trash
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server: 8.8.8.8
Address: 8.8.8.8#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
(kali㉿kali)-[~]
$
KALI

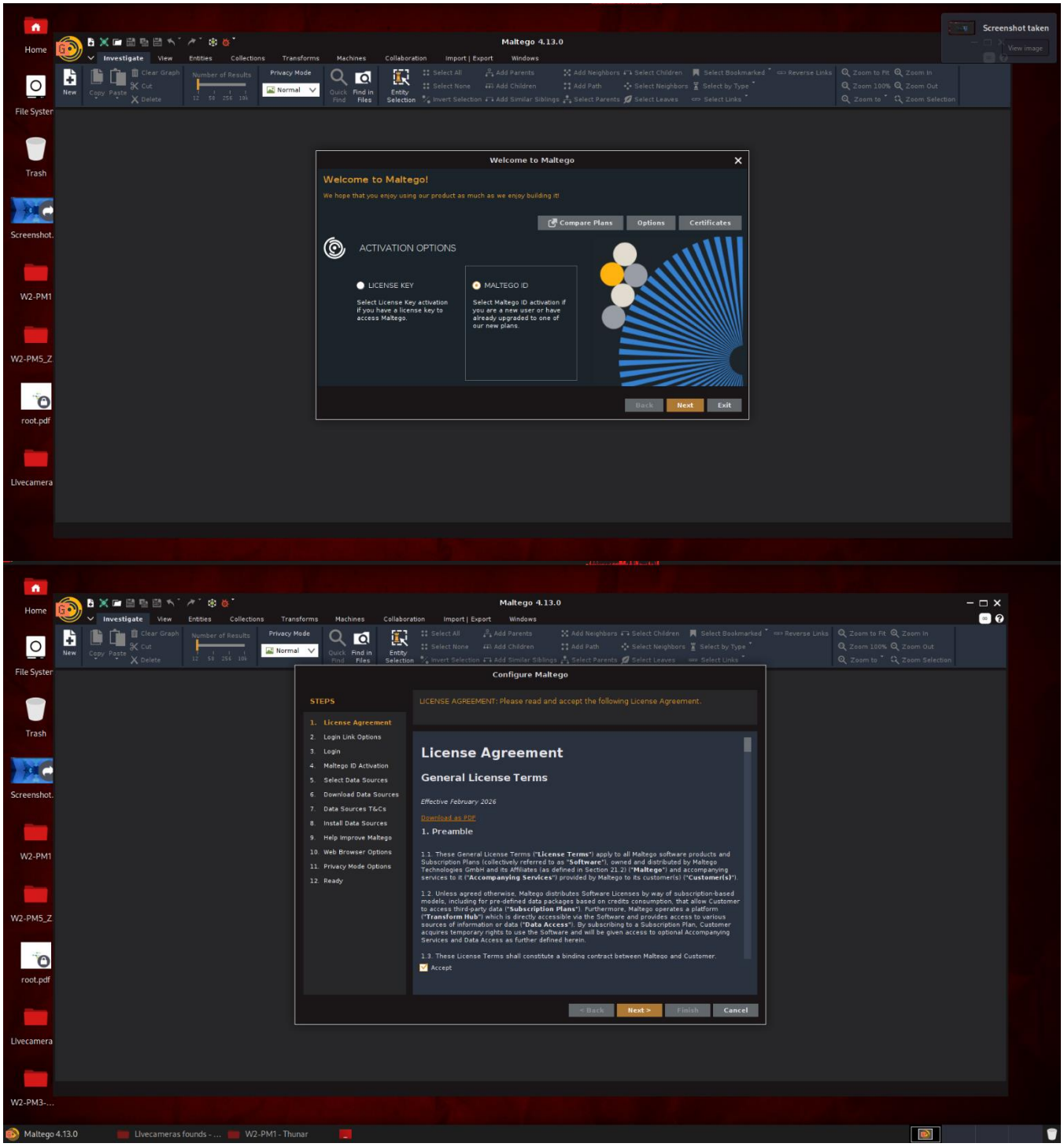

Home
File System
Trash
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=c8762ed60f81a0598c261a1407d5a093; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 14 Aug 2026 21:33:17 GMT
server: Apache
┌──(kali㉿kali)-[~]
└─$


Home
File System
Trash
sf_Downloads
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com
Woof!
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
┌──(kali㉿kali)-[~]
└─$


Applications Places
Wed Nov 11, 7:21 PM
Home
File System
Trash
Screenshot...
W2-PM1
W2-PM5_Z...
root.pdf
Zenmap
Scan Tools Profile Help
Target: 10.0.0.3/24
Profile: Intense scan, no ping
Scan
Cancel
Command: nmap -T4 -A -v -Pn 10.0.0.3/24
Hosts
Services
OS Host
10.0.0.1
10.0.0.2
10.0.0.3
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -A -v -Pn 10.0.0.3/24
Details
Nmap scan report for 10.0.0.243 [host down]
Nmap scan report for 10.0.0.244 [host down]
Nmap scan report for 10.0.0.245 [host down]
Nmap scan report for 10.0.0.246 [host down]
Nmap scan report for 10.0.0.247 [host down]
Nmap scan report for 10.0.0.248 [host down]
Nmap scan report for 10.0.0.249 [host down]
Nmap scan report for 10.0.0.250 [host down]
Nmap scan report for 10.0.0.251 [host down]
Nmap scan report for 10.0.0.252 [host down]
Nmap scan report for 10.0.0.253 [host down]
Nmap scan report for 10.0.0.254 [host down]
Nmap scan report for 10.0.0.255 [host down]
Initiating Parallel DNS resolution of 1 host. at 19:15
Completed Parallel DNS resolution of 1 host. at 19:15, 0.50s elapsed
Initiating SYN Stealth Scan at 19:15
Scanning 2 hosts [1000 ports/host]
Discovered open port 445/tcp on 10.0.0.1
Completed SYN Stealth Scan against 10.0.0.2 in 0.10s (1 host left)
Discovered open port 135/tcp on 10.0.0.1
Discovered open port 5357/tcp on 10.0.0.1
Completed SYN Stealth Scan at 19:16, 4.03s elapsed (2000 total ports)
Initiating Service scan at 19:16
Scanning 3 services on 2 hosts
Completed Service scan at 19:16, 16.79s elapsed (3 services on 2 hosts)
Initiating OS detection (try #1) against 2 hosts
Retrying OS detection (try #2) against 2 hosts
NSE: Script scanning 2 hosts.
Initiating NSE at 19:16
Completed NSE at 19:16, 7.77s elapsed
Initiating NSE at 19:16
Completed NSE at 19:16, 0.41s elapsed
Initiating NSE at 19:16
Completed NSE at 19:16, 0.01s elapsed
Nmap scan report for 10.0.0.1
Host is up (0.0016s latency).
Not shown: 997 filtered tcp ports (no-response)
Filter Hosts
Applications Places
Wed Nov 11, 6:20 PM
Home
File System
Trash
Screenshot...
W2-PM1
W2-PM5_Z...
Zenmap
Scan Tools Profile Help
Target: 10.0.0.3/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 10.0.0.3/24
Hosts
Services
OS Host
10.0.0.1
10.0.0.2
10.0.0.3
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 10.0.0.3/24
Details
Starting Nmap 7.99 (https://nmap.org) at 2026-11-11 18:20 -0500
Nmap scan report for 10.0.0.1
Host is up (0.00039s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.2
Host is up (0.00024s latency).
MAC Address: 08:00:27:D5:79:34 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.0.3
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.06 seconds
Filter Hosts
5r3
the quieter you become, the more you are able to hear


EXPLOIT DATABASE
Google Hacking Database
Filters
Reset All
Show 15
Quick Search
cam
Date Added
Dork
Category
Author
Home
Multi view
Smartphone
Gallery
Administration
Not logged in
Source 1
JavaScript
Pan, Tilt & Zoom
1
.233.191.130:8080/home.html


Maltego Graph (Desktop) 4.13.0
MALTEGO GRAPH
EVENT
Join the Maltego Community OSINT CTF
When? Friday, November 13, 15:00 CET
Calling all OSINT investigators! Join our next OSINT CTF to sharpen your analytical skills, tackle challenges with your colleagues, and have fun learning together.
Sign up now ›
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Learn how the new Maltego Transforms SDK makes it easier to connect internal, proprietary, or specialized data to Maltego Graph and build custom Transforms for your investigations!
Watch ›
WEEKLY MALTEGO STANDARD LIVE TRAINING
Join the Maltego Academy Live Sessions in English and Spanish
Grow your Maltego knowledge and expand your investigative skill set. Live sessions give you an opportunity to learn from one of our Subject Matter Experts in an engaging, interactive environment.
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GET STARTED
Maltego Data Hub
MALTEGO DATA HUB PARTNERS
Person of Interest
Cyber Threat Intelli...
Cryptocurrency
Corporate Intelligen...
Utilities
GeoSpy
alphaMountain
Farsight DNSDB
Scamadviser
AlienVault OTX
Hybrid-Analysis
HYAS - Insight
Maltego Graph (Desktop) 4.13.0
Entity Palette
New Graph (1) *
networkwalks.com
info@networkwalks.com
Overview
Detail View
Property View
Output - Transform Output


Maltego 4.13.0
Welcome to Maltego
Welcome to Maltego!
We hope that you enjoy using our product as much as we enjoy building it!
Compare Plans
Options
Certificates
ACTIVATION OPTIONS
LICENSE KEY
Select License Key activation if you have a license key to access Maltego.
MALTEGO ID
Select Maltego ID activation if you are a new user or have already upgraded to one of our new plans.
Back
Next
Exit
Configure Maltego
STEPS
1. License Agreement
2. Login Link Options
3. Login
4. Maltego ID Activation
5. Select Data Sources
6. Download Data Sources
7. Data Sources T&Cs
8. Install Data Sources
9. Help Improve Maltego
10. Web Browser Options
11. Privacy Mode Options
12. Ready
LICENSE AGREEMENT: Please read and accept the following License Agreement.
License Agreement
General License Terms
Effective February 2026
Download as PDF
1. Preamble
Accept
< Back
Next >
Finish
Cancel

-End-

👤 Author

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Cybersecurity Professional trainee B084
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Post linked in w2: https://lnkd.in/p/eYznfXvw

📌 Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub